

Another Upper Bound on the Yield Recurrence

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Let us recall the yield recurrence

$$Y_{n+2} \stackrel{d}{=} I^{(n)} \left(Y_{U_n}^{(1)} + Y_{n+1-U_n}^{(2)} + 1 \right) + J^{(n)} \max \left(Y_{U_n}^{(1)}, Y_{n+1-U_n}^{(2)} \right), \forall n \geq 1, \quad Y_1 = Y_2 = 0.$$

1.1 Escaping from the Random Variables

Let $Z_n = \lambda^{Y_n}$, where $\lambda > 0$ is a constant to be determined later. For every $I \in \{0, 1\}$ and $C \in \mathbb{R}$, we have

$$C^I = I \cdot C + (1 - I).$$

Since $I^{(n)} J^{(n)} = 0$, we have

$$\begin{aligned} & \lambda^{Y_{n+2}} \\ &= \left(I^{(n)} \left(\lambda^{Y_{U_n}^{(1)} + Y_{n+1-U_n}^{(2)} + 1} \right) + 1 - I^{(n)} \right) \left(J^{(n)} \left(\lambda^{\max(Y_{U_n}^{(1)}, Y_{n+1-U_n}^{(2)})} \right) + 1 - J^{(n)} \right) \\ &= I^{(n)} \left(\lambda^{Y_{U_n}^{(1)} + Y_{n+1-U_n}^{(2)} + 1} \right) + J^{(n)} \left(\lambda^{\max(Y_{U_n}^{(1)}, Y_{n+1-U_n}^{(2)})} \right) + 1 - J^{(n)} - I^{(n)} \\ &\leq I^{(n)} \left(\lambda^{Y_{U_n}^{(1)} + Y_{n+1-U_n}^{(2)} + 1} \right) + J^{(n)} \left(\lambda^{Y_{U_n}^{(1)}} + \lambda^{Y_{n+1-U_n}^{(2)}} \right) + 1 - I^{(n)} - J^{(n)} \end{aligned}$$

Let $z_n = \mathbb{E}[Z_n]$. Taking the expectation, we get

$$z_{n+2} \leq \frac{\lambda u}{n} \sum_{i=1}^n z_i z_{n+1-i} + \frac{2v}{n} \sum_{i=1}^n z_i + 1 - u - v.$$

Hence let us consider the sequence

$$n z_{n+2} = \lambda u \sum_{i=1}^n z_i z_{n+1-i} + 2v \sum_{i=1}^n z_i + n(1 - u - v), \forall n \geq 1, \quad z_1 = z_2 = 1. \quad (1.1)$$

Experiments show that (z_n) grows exponentially, the bound fails. That means, using $\max(x, y) \leq x + y$ keeps the sequence sublinear, but using $\lambda^{\max(x, y)} \leq \lambda^x + \lambda^y$ pushes it to linear.

1.2 Using Techniques from Analytic Combinatorics

Let $Z(t) = \sum_{n=1}^{\infty} z_n t^n$ be the generating function for the sequence $\{z_n\}$. Let us analyze each term in (1.1) through Z .

$$\begin{aligned} \sum_{n=1}^{\infty} n z_{n+2} t^n &= t \frac{d}{dt} \left(\frac{Z(t) - z_1 t - z_2 t^2}{t^2} \right) = t \frac{d}{dt} \left(\frac{Z(t)}{t^2} - \frac{1}{t} - 1 \right) = \frac{Z'(t)}{t} - \frac{2Z(t)}{t^2} + \frac{1}{t}, \\ \sum_{n=1}^{\infty} \sum_{i=1}^n z_i z_{n+1-i} t^n &= \frac{Z(t)^2}{t}, \\ \sum_{n=1}^{\infty} \sum_{i=1}^n z_i t^n &= \frac{Z(t)}{1-t}, \\ \sum_{n=1}^{\infty} n t^n &= \frac{t}{(1-t)^2}. \end{aligned}$$

The function equation is

$$\frac{Z'(t)}{t} - \frac{2Z(t)}{t^2} + \frac{1}{t} = \lambda u \frac{Z(t)^2}{t} + 2v \frac{Z(t)}{1-t} + (1-u-v) \frac{t}{(1-t)^2}.$$

Equivalently,

$$Z'(t) = \lambda u Z(t)^2 + \left(\frac{2}{t} + \frac{2vt}{1-t} \right) Z(t) + \frac{(1-u-v)t^2}{(1-t)^2} - 1. \quad (1.2)$$

Substituting $Z(t) = -\frac{1}{\lambda u} \frac{W'(t)}{W(t)}$ in the convention of a Riccati equation, we get

$$\begin{aligned} -\frac{1}{\lambda u} \frac{W''(t)W(t) - W'(t)^2}{W(t)^2} &= \frac{1}{\lambda u} \frac{W'(t)^2}{W(t)^2} - \frac{1}{\lambda u} \left(\frac{2}{t} + \frac{2vt}{1-t} \right) \frac{W'(t)}{W(t)} + \frac{(1-u-v)t^2}{(1-t)^2} - 1 \\ \iff -\frac{1}{\lambda u} \frac{W''(t)}{W(t)} &= -\frac{1}{\lambda u} \left(\frac{2}{t} + \frac{2vt}{1-t} \right) \frac{W'(t)}{W(t)} + \frac{(1-u-v)t^2}{(1-t)^2} - 1 \\ \iff W''(t) - \left(\frac{2}{t} + \frac{2vt}{1-t} \right) W'(t) &+ \left(\frac{(1-u-v)t^2}{(1-t)^2} - 1 \right) \lambda u W(t) = 0. \end{aligned}$$

Generating Random Regex Tree Uniformly

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The goal is to sample recursively, avoid listing all trees. We need to count the number of trees whose root is either $\cdot$, $+$, or $*$ in order to determine the probabilities of choosing each operator at the root. Suppose that all leaves are the same.

2.1 Deriving the Recurrences

Let t_n be the number of regex trees with n leaves that is not a trivial leaf. Let t_n^+ and t_n^* be the number of regex trees with n leaves whose root is $+$, and $*$ respectively. Then

$$t_n = 2t_n^+ + t_n^* \text{ for } n \geq 2, \quad (2.1)$$

$$t_n^* = t_{n-1} \text{ for } n \geq 2, \quad (2.2)$$

$$t_n^+ = \sum_{i=1}^{n-2} t_i t_{n-1-i} \text{ for } n \geq 3, \quad (2.3)$$

$$t_1^+ = t_1^* = 0, t_2^* = 0, t_2^+ = 1, t_1 = 1. \quad (2.4)$$

Using generating functions $T_n(z) = \sum_{k=0}^n t_k z^k$ (similarly for T_n^+ and T_n^*), we have

$$\begin{aligned} T^*(x) &= xT(x) \\ T^+(x) &= x(T(x))^2 \\ T(x) &= x + 2T^+(x) + T^*(x) \end{aligned}$$

Solving for T , we get

$$T(x) = x + 2(x(T(x))^2) + (xT(x)).$$

We choose the root where $T(0) = 0$.

$$T(x) = \frac{1 - x - \sqrt{1 - 2x - 7x^2}}{4x}. \quad (2.5)$$

Let $\Delta = 1 - 2x - 7x^2$. Then $\Delta' = -2 - 14x$. We have

$$\begin{aligned} 4xT(x) &= 1 - x - \sqrt{\Delta} \\ \iff 4T(x) + 4xT'(x) &= -1 + \frac{\Delta'}{2\sqrt{\Delta}} \\ \iff 4T(x) + 4xT'(x) &= -1 + \frac{\Delta'\sqrt{\Delta}}{2\Delta} \\ \iff 8\Delta T(x) + 8\Delta xT'(x) &= -2\Delta + \Delta'(4xT(x) - 1 + x) \\ \iff x\Delta T'(x) + \left(\Delta - \frac{1}{2}x\Delta'\right)T(x) &= \frac{(x-1)\Delta' - 2\Delta}{8} \\ \iff x(1 - 2x - 7x^2)T'(x) + (1-x)T(x) &= 2x. \end{aligned}$$

Identify the coefficients of x^n with $n \geq 2$ to 0, we get

$$\begin{aligned} nt_n - 2(n-1)t_{n-1} - 7(n-2)t_{n-2} + t_n - t_{n-1} &= 0 \\ \iff (n+1)t_n &= (2n-1)t_{n-1} + 7(n-2)t_{n-2} \end{aligned} \quad (2.6)$$

Given that $t_1 = t_2 = 1$, we can compute t_n in linear time.

2.2 Sampling

We have

$$\sum_{i=1}^{n-2} t_i t_{n-1-i} = t_n^+ = \frac{1}{2}(t_n - t_n^*) = \frac{1}{2}(t_n - t_{n-1}) \text{ for } n \geq 3.$$

The probabilities of the root are

$$p_n(\cdot) = \frac{t_{n-1}}{t_n} \text{ and } p_n(+) = p_n(\cdot) = \frac{t_n^+}{t_n} = \frac{t_n - t_{n-1}}{2t_n}. \quad (2.7)$$

The remaining task is to ensure that the way subtrees are chosen preserves the uniform distribution. We can sample an integer uniformly in $\left\{1, \dots, \frac{1}{2}(t_n - t_{n-1})\right\}$, regard each term $t_i t_{n-1-i}$ as a block, and determine which block the integer falls into. An optimized way is to order them as $1, n, 2, n-1, \dots$